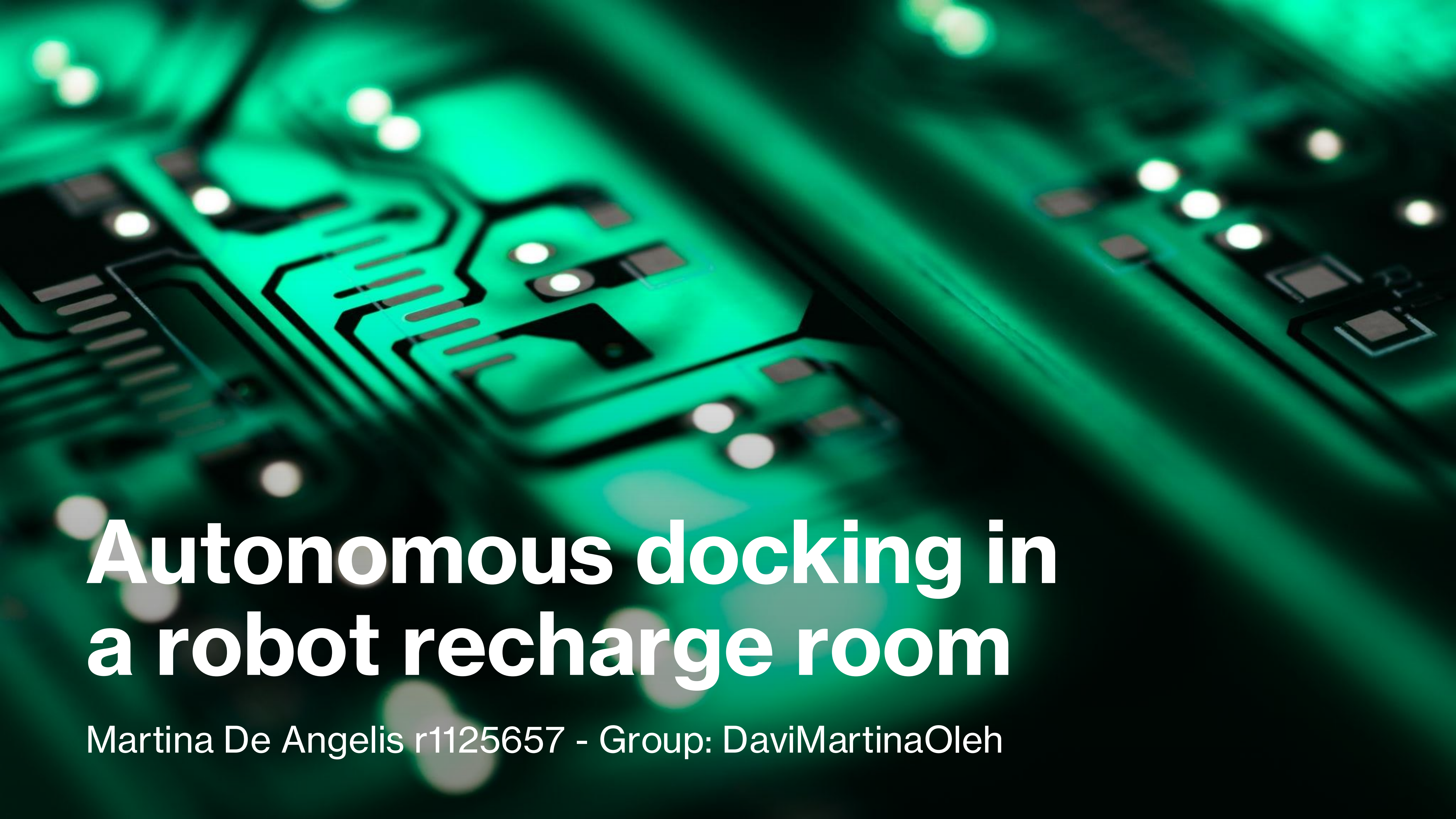

Autonomous docking in a robot recharge room

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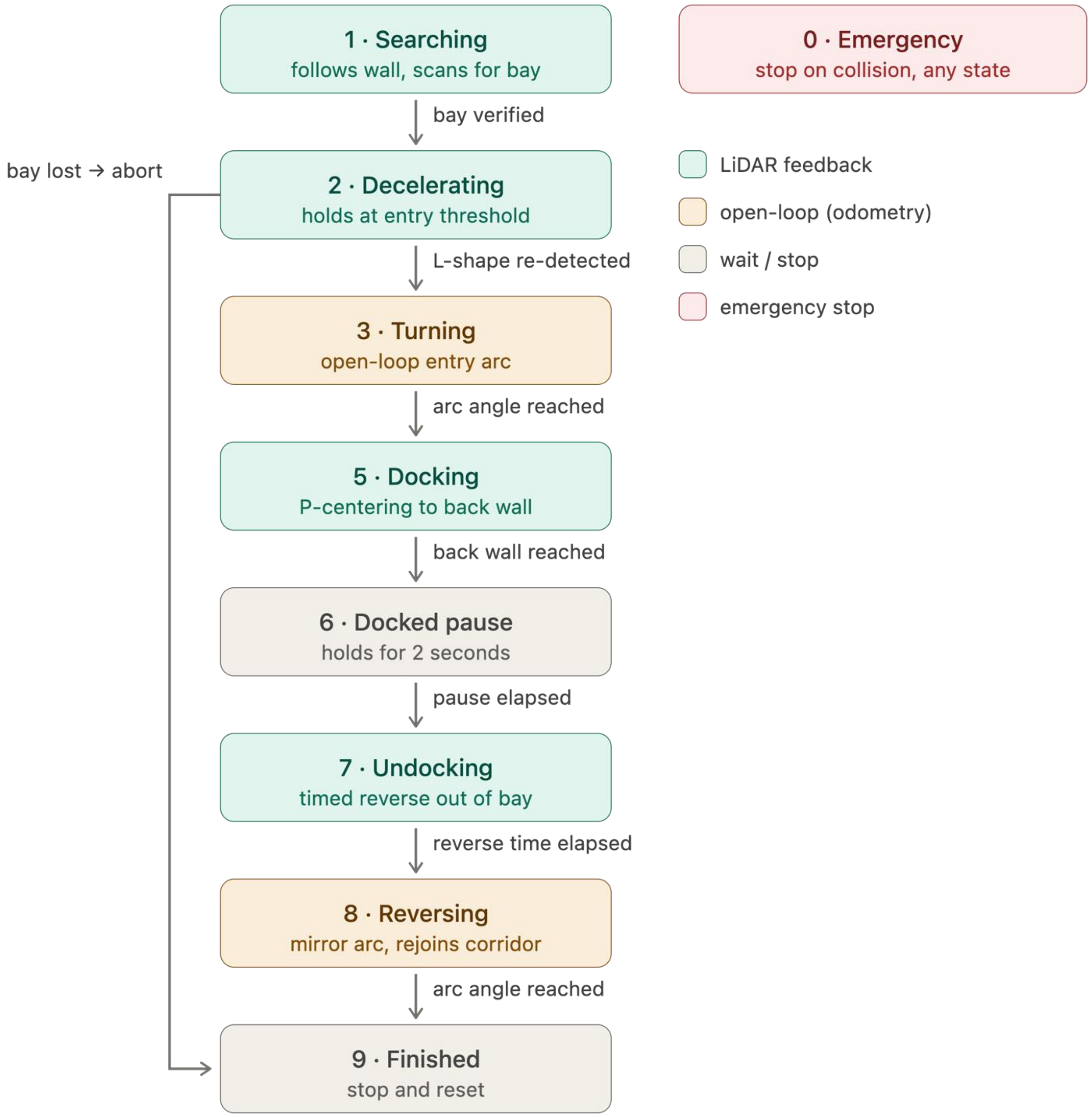
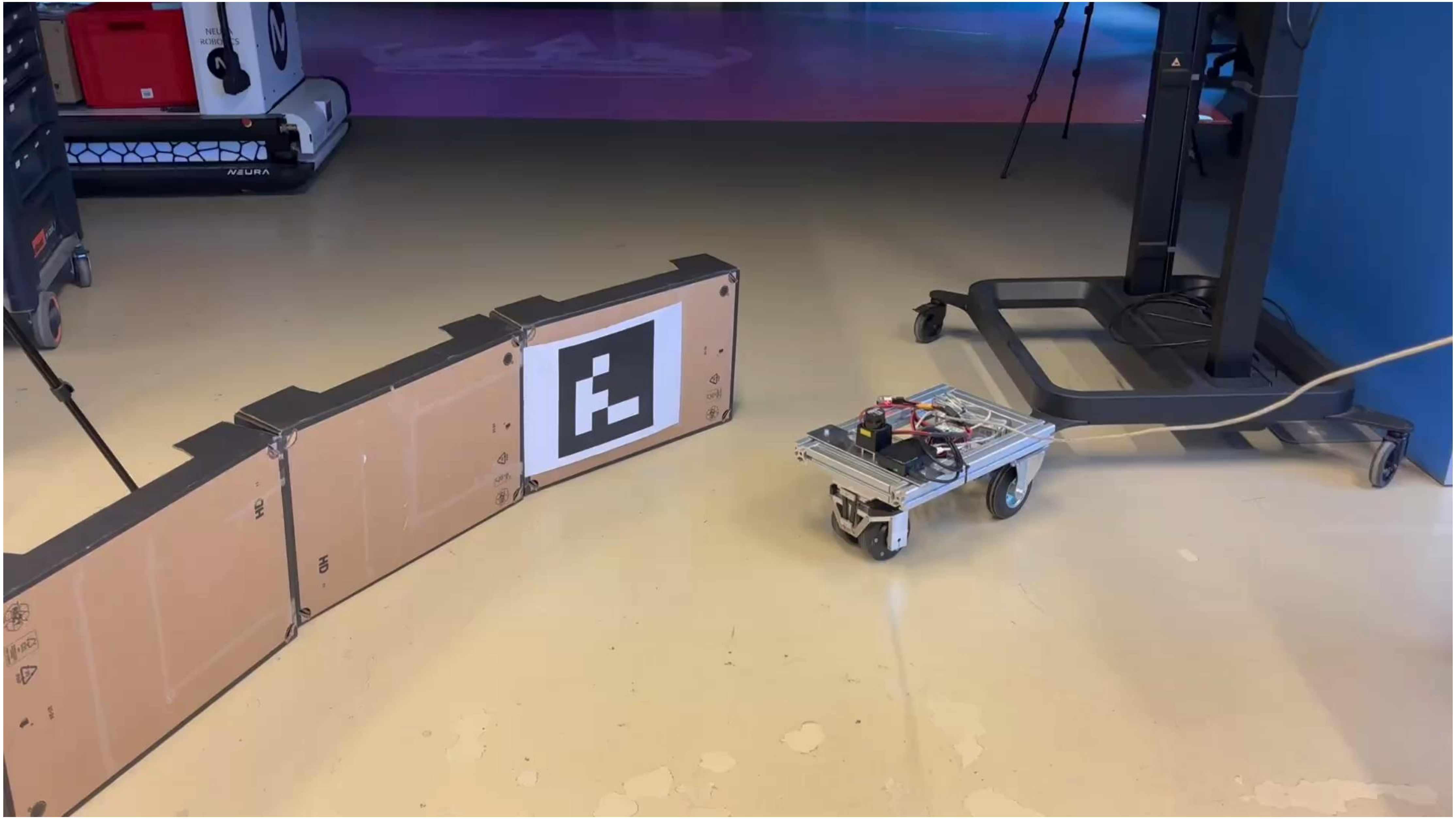
Updated FSM



1 · Searching
follows wall, scans for bay
0 · Emergency
stop on collision, any state
bay verified
bay lost → abort
2 · Decelerating
holds at entry threshold
LiDAR feedback
open-loop (odometry)
wait / stop
emergency stop
L-shape re-detected
3 · Turning
open-loop entry arc
arc angle reached
5 · Docking
P-centering to back wall
back wall reached
6 · Docked pause
holds for 2 seconds
pause elapsed
7 · Undocking
timed reverse out of bay
reverse time elapsed
8 · Reversing
mirror arc, rejoins corridor
arc angle reached
9 · Finished
stop and reset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Updates

Bay estimation and tracking (State 1)

- Report: single gating radius + dedicated bay frame
- Unexpected: one gate either lost the bay or accepted false detections; bay frame unnecessary
- Fix: predictor-corrector in robot frame (odometry predicts, LiDAR corrects) with phase-dependent gates; odometry-only once anchors leave view

Deceleration and entry positioning (State 2)

- Report: PID braking + gain-scheduled LQR
- Unexpected: needless at low speed, added tuning complexity
- Fix: threshold stop at near-anchor $x = 0.40$ m + L-shape re-validation

Entry arc (State 3)

- Report: feedforward arc with closed-loop path feedback
- Unexpected: anchors leave LiDAR field of view during the turn
- Fix: open-loop arc on odometry yaw; radius sized from bay width, clamped to platform limits

Updates

Final docking (State 4/5)

- Report: full PID centering, stop at offset
- Unexpected: I/D terms unnecessary; terminal pose better fixed geometrically
- Fix: proportional-only control; stop when front cone (±5°) sees front wall at 15 cm

Undocking and exit (State 5)

- Report: mirrored arc with negated steering angle, relying on odometry
- Unexpected: anchors behind robot, exit geometry unreliable
- Fix: reverse for same duration as approach (timing symmetry) + open-loop mirror arc; low speed, large margins

Failure case:
short bay leaves no room to realign